OPENROCKET ROCKET DESIGN

Aim:

Design, simulate, and optimize a single-stage model rocket using OpenRocket based on the specifications and constraints provided below.

1. Mission Requirements:

• Target Apogee: Minimum 1000 meters

A target apogee of 1056 m has been achieved.

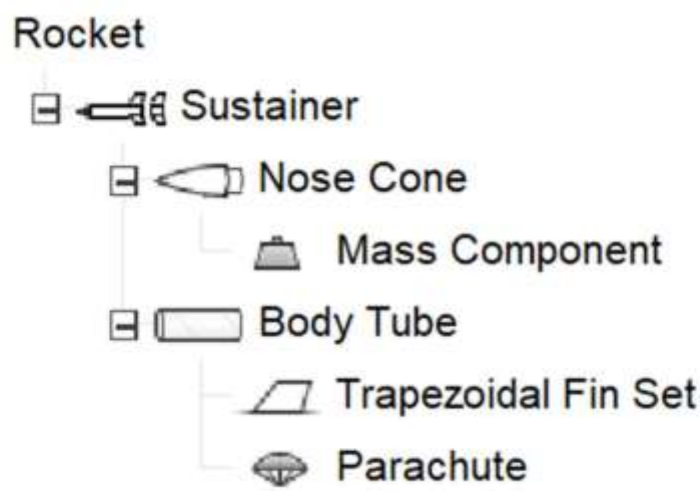
• Stable Flight: Stability margin between 1.0–2.0 calibers

Stability= 1.52 cal

• Safe Recovery: Descent rate less than10m/s

Ground hit velocity= 5.11m/s

• Single-stage rocket configuration



2. Design Constraints

• Total Length: Between 800mm and 1200mm

The length is 1200mm

• Body Diameter:54mm (fixed)

• Maximum Lift-o Mass: 1.5kg

Lift-o mass taken is 734g

• Material:

Carboard body tube and fiberglass fins

• Maximum Mach Number: Below 0.8 (subsonic preferred)

0.702

Subsystem Requirements

1. Nose Cone:

• Choose between Ogive, Elliptical, or Conical

An ogive nose cone has been chosen

• Justification of the selection based on aerodynamics

An ogive nose cone is selected as its providing a smooth aerodynamic profile that minimizes drag during subsonic and transonic flight. The curved shape allows airflow to remain attached to the surface therefore, reducing pressure drag and rocket’s e iciency and stability is improved.

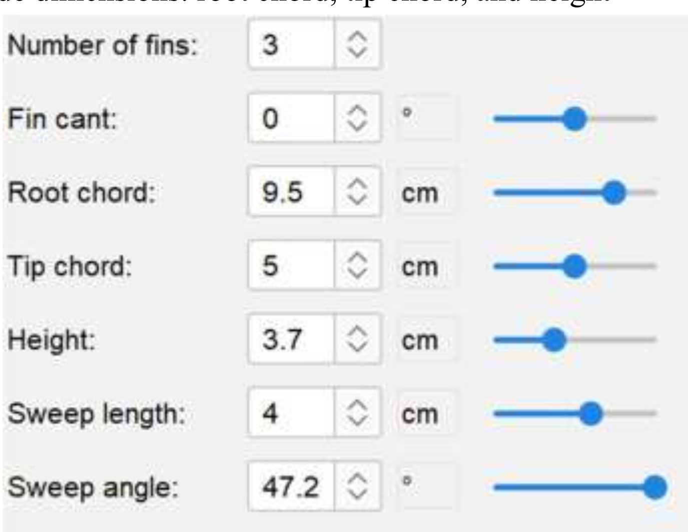
2. Fins:

Trapezoidal fins were selected because a good balance between the aerodynamic stability and structural simplicity. The fin dimensions are optimized to make sure that the centre of pressure remains behind the centre of gravity, thereby achieving the required stability margin between 1 and 2.

• Minimum 3 fins (Trapezoidal or Elliptical)

Traperzoidal finsets were chosen

• Provide dimensions: root chord, tip chord, and height



CP shifts appropriately for required stability

3. Propulsion:

- Use a certified commercial motor (H-class recommended) □

[HP-H135W-10]

- Thrust-to-weight ratio must be greater than 5:1 at lift-o   Avionics & Payload: □ □ Include a minimum payload mass of 75 grams Include an altimeter for apogee detection

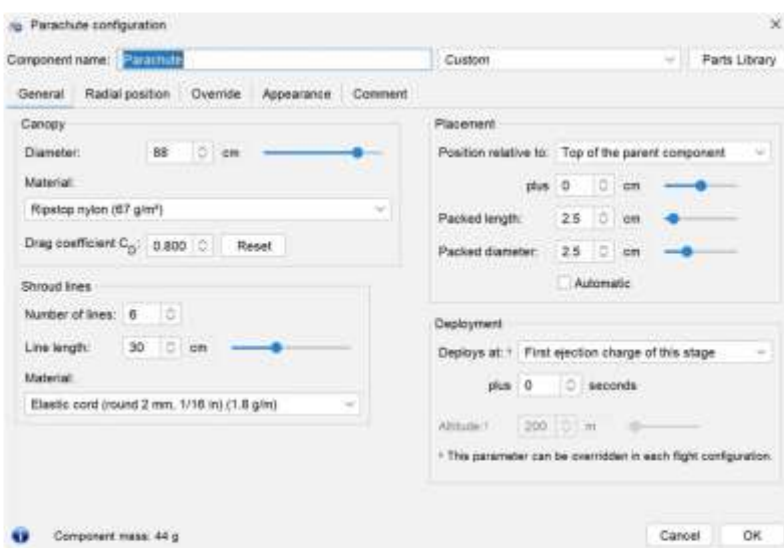
The HP-H135W commercial rocket motor is selected for propulsion. The H-class motor provides su icient thrust to achieve the target apogee while maintain a trustto-weight ratio greater than 5:1 at lift-o   . We have a ratio of 27:1. The motor also ensures that the rocket remains within the required Mach number limit of 0.8. A ratio of 0.702 was achieved.

4. Recovery System:

A parachute recovery system was implemented to ensure safe descent of the rocket after apogee. A shock cord and recovery harness connects the nose cone and body tube to absorb the ejection shock and prevent structural damage during deployment.

- Parachute or dual- deployment system

- Shock cord and recovery harness included



Performance Verification (Simulation Output Required) Students must submits creenshots showing:

- Apogee(≥1000m)

- Stability Margin (1.0– 2.0 calibers)
- Maximum Velocity
- Acceleration Profile
- CG and CP locations

Flight configuration: [HP-H135W-10]
Apogee: 1056 m
Max. velocity: 238 m/s (Mach 0.702)
Max. acceleration: 208 m/s²

Stability: 1.52 cal / 6.83 %
CG: 68.7 cm
CP: 76.9 cm

5. Avionics and Payload:
A payload mass of 75g is included in the nose section to represent onboard avionics such as altimeter. The altimeter is used to detect the apogee of the rocket and trigger the recovery system deployment.

Bonus Challenge (Optional) : Optimize your rocket to reach the highest possible altitude while maintaining stability within the safe range and keeping Mach number below 0.8

The optimization of the rocket geometry and mass distribution was done through simulation in OpenRocket to maximize altitude while maintain the required stability margin and keeping the max Mach number below 0.8.

Maintaining a Mach number below 0.8, we have obtained a maximum altitude of 1056m.

Flight simulation results:

Rocket design | Motors & Configuration | Flight simulations

New simulation | Edit simulation | Run simulations | Delete simulations | Plot / Export

Wa...	Name	Configuration	Velocity off rod	Apogee	Velocity at de...	Optimum delay	Max. velocity	Max. accelera...	Time to apogee	Flight time	Ground hit vel...
✓	Simulation 1	[HP-H135W-10]	18.7 m/s	1056 m	3.36 m/s	10.3 s	238 m/s	208 m/s²	12.3 s	235 s	5.11 m/s

The rocket design was simulated in OpenRocket to evaluate its flight performance. The rocket achieved a maximum apogee of 1056 m, meeting the minimum requirement of 1000m. the maximum velocity was 238 m/s (0.70), which is below the limit of Mach 0.8. The parachute recovery system ensures safe landing with a ground hit velocity of 5.11 m/s, which is within the required safety limit.

Conclusion:

In this project, a single-stage model rocket was designed and simulated using OpenRocket according to the specified mission and design constraints. The rocket configuration, including the ogive nose cone, trapezoidal fins, payload mass, and parachute recovery system, was optimized to achieve stable and e icient flight performance. Simulation results demonstrate that the rocket successfully reaches an apogee above 1000 m while maintaining a stability margin within the safe range and keeping the maximum Mach number below 0.8. The recovery system ensures a safe descent with a low ground impact velocity. Overall, the project highlights how simulation tools can be e iciently used to design and evaluate rocket performance before actual implementation.